OLIST E-COMMERCE

Performance & Customer Insights

A Power BI Data Analytics Case Study

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Tools: Power BI • Power Query • DAX

1. Executive Summary

This project analyses the Olist Brazilian e-commerce dataset to demonstrate how order, product, customer, payment and review data can be organised in Power BI and translated into business questions. The report focuses on product sales, order fulfilment, customer satisfaction and repeat purchasing.

Preliminary dashboard results show that Health & Beauty is the highest-revenue category in the current view, 97.02% of orders are marked delivered, 57.78% of recorded reviews are five-star, and approximately 3.01% of unique customer identities are classified as repeat customers by the current DAX calculation. These are exploratory results and should be revalidated before formal business use.

2. Business Problem

An e-commerce business needs to understand what drives sales, whether orders reach customers, how customers rate their experience, and whether customers return to purchase again. A dashboard can summarise these dimensions, but business value comes from validating measures, identifying meaningful patterns and recommending evidence-based actions.

3. Project Objectives

- Summarise sales and order activity through key performance indicators.
- Identify product categories contributing the most to product-price revenue.
- Measure the share of orders with a delivered status.
- Assess the distribution of customer review scores.
- Estimate the share of unique customer identities with more than one order.
- Translate findings into practical questions and recommendations for business teams.

4. Dataset and Scope

The analysis uses the Olist Brazilian e-commerce public dataset, with separate tables for orders, customers, order items, payments, reviews, products, sellers and product-category translations. The geolocation table was excluded from the initial report because it is large and needs additional preparation before reliable geographic analysis.

The dataset is transactional rather than a complete profit-and-loss view. The current revenue measure sums item prices and excludes freight charges, costs, refunds and other adjustments. It is therefore described as product-price revenue or gross product sales, not net revenue or profit.

5. Tools and Techniques

- Power Query: importing tables, reviewing data types and inspecting missing values.
- Power BI data model: connecting customer, order, item, product, seller, payment, review and category tables.
- DAX: measures for product-price revenue, distinct orders, average order value, delivered orders, delivery rate, five-star reviews and repeat customers.
- Visual analytics: KPI cards, monthly trend line, category revenue chart, review-score distribution, order-status distribution and a purchase-date slicer.

6. Data Preparation and Modelling

Identifiers were treated as text, timestamps as date/time values, and numeric fields such as price and freight as numeric values. Missing values in product attributes and delivery timestamps were not automatically removed because they may represent incomplete product metadata or orders that had not reached a delivery milestone.

The model was designed around the orders table, linked to customers and to order-level or order-item-level tables. In a production-quality analysis, relationship cardinality, filter direction, duplicate keys and measure behaviour should be verified explicitly. Item, payment and review tables can contain multiple rows per order, so counts should use distinct order IDs when the intended metric is orders.

7. Measures Used

Measure	Definition	Interpretation / caveat
Total Revenue	SUM of order-item price	Item prices only; excludes freight and may include cancelled/unavailable orders depending on filters.
Total Orders	Distinct count of order_id	Counts orders in the selected filter context; status scope should be specified.
Average Order Value	Total Revenue ÷ Total Orders	Interpret cautiously until revenue and order-status scope are aligned.
Delivered Orders	Distinct order_id where status = delivered	Counts orders currently marked delivered.
Delivery Rate	Delivered Orders ÷ Total Orders	Reported as 97.02%; validate denominator and date filtering.
Five-Star Reviews	Count of review rows with score = 5	Counts review records, not necessarily unique customers.
Five-Star Review Rate	Five-star review rows ÷ all review rows	Reported as 57.78%; validate duplicates/multiple reviews per order.
Frequent Customers	Unique customer identities with >1 distinct order	Reported as 2,997; depends on correct customer-to-order filter propagation.
Frequent Customer Rate	Frequent Customers ÷ Unique Customers	Approximately 3.01% using 2,997 / 99,441; validate model and cohort scope.

8. Preliminary Findings

8.1 Product category performance

Health & Beauty ranked first in the revenue-by-category visual, followed by Watches & Gifts, Bed, Bath & Table, Sports & Leisure, and Computers & Accessories. These categories merit further investigation. Revenue rank alone does not establish profitability, demand growth or inventory efficiency.

8.2 Order fulfilment

The dashboard reports a 97.02% delivery rate. This is a useful starting point, but it is not an on-time delivery rate. A separate analysis comparing estimated and actual delivery dates is required to measure timeliness. Cancelled, unavailable and other non-delivered orders should be investigated.

8.3 Customer satisfaction

The dashboard reports that 57.78% of review records have a score of five. Five-star ratings therefore form a majority of recorded reviews, but this does not explain why customers are satisfied. Lower ratings and review comments should be examined to identify recurring delivery, product or service issues.

8.4 Repeat purchasing

The current calculation identifies 2,997 frequent customers out of 99,441 unique customer identities, or approximately 3.01%. Treat this as a preliminary estimate until the customer-to-order relationship is validated. The dataset's observation window also limits conclusions about long-term loyalty.

8.5 Revenue interpretation

The current measure sums item prices and excludes freight. It may include items attached to cancelled or unavailable orders, depending on filters. Before presenting revenue or AOV as formal KPIs, define eligible order statuses and ensure numerator and denominator use a consistent cohort.

9. Business Recommendations

1. **Investigate leading categories:** Compare category sales over time, order volume, average item price and product-level concentration before making inventory or marketing decisions.
2. **Improve repeat purchasing:** Test post-purchase messages, targeted recommendations and loyalty incentives; evaluate results using repeat-purchase cohorts.
3. **Learn from lower reviews:** Segment ratings by category, seller, delivery performance and review comments to locate recurring pain points.
4. **Investigate non-delivered orders:** Review cancelled and unavailable orders by seller, category and purchase period; quantify their contribution before prioritising changes.
5. **Measure delivery timeliness separately:** Compare actual delivery timestamps with estimated dates and report on-time delivery rate, late-delivery days and missing timestamps.
6. **Strengthen KPI definitions:** Document whether each measure includes freight, which statuses qualify and whether it counts orders, items, review records or unique customers.

10. Limitations

- The results are based on values reported from the current Power BI report and have not been independently recalculated from source CSV files.
- The revenue measure is item-price revenue, not profit or necessarily realised net revenue.
- Delivery rate is not the same as on-time delivery rate.
- Repeat-customer results depend on model relationships, identity definitions and the observation period.
- Review records may not map one-to-one to unique orders or customers.
- The analysis is descriptive and does not establish causal effects.

11. Conclusion

This project demonstrates a practical Power BI workflow: combining multiple e-commerce tables, creating DAX measures, building interactive visuals and interpreting results in business terms. Preliminary findings highlight category revenue concentration, a high share of delivered-status orders, a majority of five-star reviews and a possible repeat-purchase opportunity.